Charles Grassi

Graphics Programmer & Technical Artist

Zylaris Ltd - Cyprus
contact.charles.grassi@gmail.com
(+33) 6 46 11 82 14
linkedin.com/in/charles-grassi
charlesgrassi.dev

PROFESSIONAL SUMMARY

Graphics programmer with **7+ years of experience** specializing in real-time rendering, shader development, and GPU optimization. Expert in Unity, Unreal Engine, WebGPU and low-level graphics APIs. Also builds AI agents and Python backends: CTO of Events Arena. Shipped **100+ production shaders** and optimized rendering pipelines reaching **2M+ end users**.

70%	100+	2M+	15x	12+
AVG PERF GAIN	SHADERS	END USERS	FASTER BAKE	STUDIOS

PROFESSIONAL EXPERIENCE

Chief Technology Officer — 2025-now
Events Arena

Built the company's operating platform: booking intake, seat allocation, pricing, ticket logistics. **1,000+ commits**, **80K lines of Python**, Claude agent with **65 tools** in production.

`FastAPI` `PostgreSQL` `AI Agents`

Co-founder & Graphics Lead — 2026
HYPERDRYFT

Live-visuals engine: node graphs compile to **multipass WGSL on WebGPU**, audio-reactive. **300+ nodes**, AI copilot that writes shaders. Rust/wgpu desktop + web.

`WebGPU` `Rust` `AI`

Technical Artist & Graphics Programmer — 2025
Nexus Studios

Built real-time GI system using SDF for VR, achieving **90 FPS on Quest 2** with **20+ lights**. Reduced baking from **45min to 3min (15x)**.

`Compute` `SDF` `VR` `GI`

Chief Technology Officer — 2023-24
SABDA

Led **16K 360° immersive installation** across **6 servers at 90 FPS**. Managed **64-ch spatial audio**, **2.5GB/s throughput**. Led **team of 8**.

`Projection` `Audio` `Leadership`

Lead Graphics Programmer — 2023
Little Red Dog Games

Led **4 engineers** developing VFX with compute shaders. **70% CPU improvement** (12ms→3.6ms). Shipped **15+ effects**.

`Compute` `Burst` `VFX`

Senior Graphics Programmer — 2023
22cans

Created **25+ material shader library**. Grass system with **1M+ blades at 60 FPS mobile**. Volumetrics with **sub-5ms GPU**.

`HLSL` `Mobile` `Volumetrics`

Graphics Programmer — 2021-22
Scapin' / The Sandbox

Custom lighting for **30+ lights** reaching **2M+ users**. Optimized blur **18ms→4ms**. Built 8-layer parallax clouds.

`Lighting` `Mobile` `Web3`

Graphics Programmer — 2020-21
Ubisoft

AAA development: **20+ shaders** for PC/PS5/Xbox. **50% GPU reduction** while **2x particle count to 100K+** at **4K 60 FPS**.

`AAA` `VFX` `Console`

TECHNICAL SKILLS

CORE
Unity, Unreal, C#, C++, Rust, Python, HLSL, GLSL

GRAPHICS APIS
WebGPU/WGSL, DX11/12, Vulkan, OpenGL, WebGL, Metal

UNITY
URP, HDRP, VFX Graph, Shader Graph

SPECIALIZATIONS
AR/VR, Compute, Niagara, GI, PBR

TOOLS
RenderDoc, NSight, PIX, Frame Debugger

LANGUAGES
English (Fluent), French (Native)

CLIENTS

Events Arena · Nexus Studios · Ubisoft · 22cans · The Sandbox · DREST · H3 Space · Little Red Dog · SABDA · GN3RA · Lunchbox Ent.

Portfolio: Interactive demos & shader breakdowns at charlesgrassi.dev | artstation.com/charlesgrassi